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3. Deplasmolysis

The effect of plasmolysis can be reversed, by transferring them back into water or **hypotonic solution**. Due to endosmosis, the cell becomes turgid again. It regains its original shape and size. This phenomenon of the revival of the plasmolysed cell is called **deplasmolysis**. Example: Immersion of dry raisin in water.

Potato Osmoscope

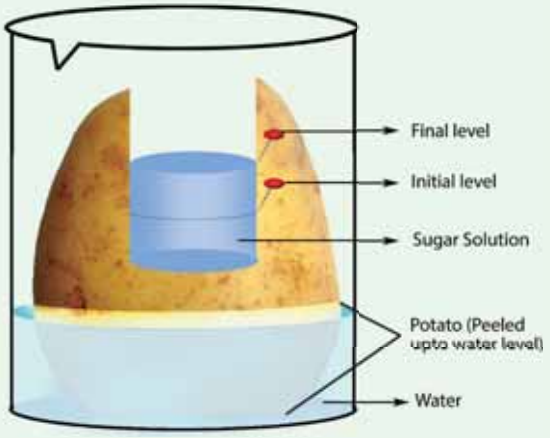


Figure 11.8: Demonstration of Endosmosis by Potato Osmoscope

i. Take a peeled potato tuber and make a cavity inside with the help of a knife.
ii. Fill the cavity with concentrated sugar solution and mark the initial level.
iii. Place this setup in a beaker of pure water.
iv. After 10 minutes observe the sugar solution level and record your findings (Figure 11.8).
v. With the help of your teacher discuss the results.

Instead of potato use beetroot or bottle-guard and repeat the above experiment. Compare and discuss the results.

4. Reverse Osmosis

Reverse Osmosis follows the same principles of osmosis, but in the reverse direction. In this process movement of water is reversed by applying pressure to force the water against a concentration gradient of the solution. In regular osmosis, the water molecules move from the higher concentration (pure water = hypotonic) to lower concentration (salt water = hypertonic). But in reverse osmosis, the water molecules move from the lower concentration (salt water = hypertonic) to higher concentration (pure water = hypotonic) through a selectively permeable membrane (Figure 11.9).

Uses: Reverse osmosis is used for purification of drinking water and desalination of seawater.

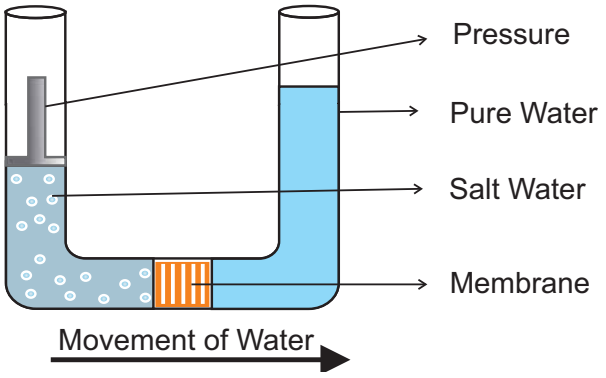


Figure 11.9: Reverse Osmosis

Check your grasp!

If a cell in the cortex with DPD of 5atm is surrounded by hypodermal cells with DPD of 2atm, what will be direction of movement of water?

Solution: Water will move from low DPD to high DPD (hypodermis 2 atm to cortex 5 atm).